

```
!pip install --upgrade diffusers transformers -q
```

```
from pathlib import Path
import tqdm
import torch
import pandas as pd
import numpy as np
from diffusers import StableDiffusionPipeline
from transformers import pipeline, set_seed
import matplotlib.pyplot as plt
import matplotlib.pyplot as plt
import cv2
```

```
import torch
class CFG:
    device = "cuda"
    seed = 42
    image_gen_steps = 35
    image_gen_model_id = "stabilityai/stable-diffusion-2"
    image_gen_size = (700, 600)
    image_gen_guidance_scale = 9
    prompt_gen_model_id = "gpt3"
    prompt_dataset_size = 6
    prompt_max_length = 12

    @classmethod
    def initialize_generator(cls):
        if cls.device == "cuda" and torch.cuda.is_available():
            cls.generator = torch.Generator(device='cuda').manual_seed(cls.seed)
        else:
            cls.generator = torch.Generator().manual_seed(cls.seed)

# Initialize the generator
CFG.initialize_generator()
```

```
import torch

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
image_gen_model = StableDiffusionPipeline.from_pretrained(
    CFG.image_gen_model_id,
    torch_dtype=torch.float,
    revision="fp16",
    use_auth_token='your_hugging_face_auth_token',
    guidance_scale=9
)

# Move the model to the device
image_gen_model = image_gen_model.to(device)
```

```
/usr/local/lib/python3.12/dist-packages/diffusers/pipelines/pipeline_loading_utils.py:333: FutureWarning: You are loading tf
The Diffusers team and community would be very grateful if you could open an issue: https://github.com/huggingface/diffuser
warnings.warn(
Keyword arguments {'use_auth_token': 'your_hugging_face_auth_token', 'guidance_scale': 9} are not expected by StableDiffusio
Loading pipeline components...: 100% 5/5 [00:08<00:00, 2.21s/it]
An error occurred while trying to fetch /root/.cache/huggingface/hub/models--stabilityai--stable-diffusion-2/snapshots/d75b6
Defaulting to unsafe serialization. Pass `allow_pickle=False` to raise an error instead.
An error occurred while trying to fetch /root/.cache/huggingface/hub/models--stabilityai--stable-diffusion-2/snapshots/d75b6
Defaulting to unsafe serialization. Pass `allow_pickle=False` to raise an error instead.
```

```
def generate_image(prompt, model):
    image = model(
        prompt, num_inference_steps=CFG.image_gen_steps,
        generator=CFG.generator,
        guidance_scale=CFG.image_gen_guidance_scale
    ).images[0]

    image = image.resize(CFG.image_gen_size)
    return image
```

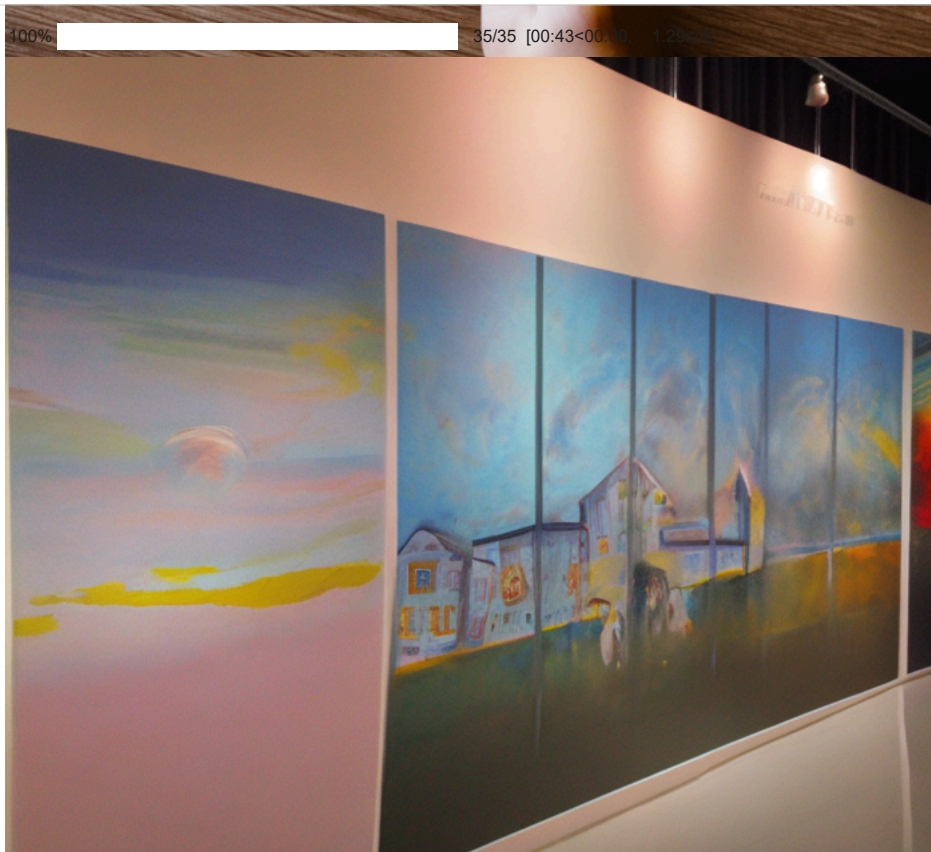
```
generate_image("coffee with pen and laptop at side", image_gen_model)
```

100%

35/35 [00:43<00:00, 1.26s/it]



```
generate_image("Painting Exhibition background", image_gen_model)
```



```
generate_image("Children playing football in the ground in colorful background", image_gen_model)
```

100%

35/35 [00:44<00:00, 1.27s/it]



```
generate_image("Realistic astronaut floating in space, detailed suit, Earth in the background, high-definition.", image_gen_
```



Start coding or [generate](#) with AI.